Model Predictive Control

Chapter 1: Introduction to MPC

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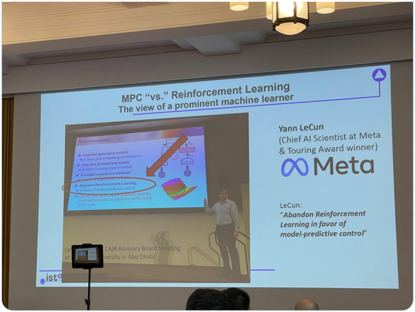
Model Predictive Control vs. Reinforcement Learning - Snapshots from the Web

Yann LeCun @ylecun

Indeed, I do favor MPC over RL.
I've been making that point since at least 2016.
RL requires ridiculously large numbers of trials to learn any new task.
In contrast MPC is zero shot: If you have a good world model and a good task objective, MPC can solve new tasks without any task-specific learning.
That's the magic of planning.
It doesn't mean that RL is useless, but its use should be a last resort.

> **Antoine Leeman** @antoine_leeman · 24 Aug 2024
> Yann Lecun favours MPC over RL? Interesting! What about data-augmented MPC? @ylecun #nmpc2024



9:17 pm · 25 Aug 2024 · **265.5K** Views

Dimitri Bertsekas · 1st
Fulton Professor at School of Computing and Augmented Intelligence, Ar...
2mo · Edited ·

I recently came across the statement "Abandon Reinforcement Learning in favor of model predictive control," by Turing Award winner and deep learning pioneer Yann LeCun from one of his video lectures ("Objective-Driven AI" talk at the University of Washington, 2024). I found it remarkable and courageous, as it challenges the prevailing orthodoxy - at least outside the decision and control community. LeCun expands on this view in several public lectures.

The statement resonated with me, as I have expressed a related perspective in my books and in my course at ASU since 2019.
In summary: *MPC is part of RL, and reversely, the most reliable part of RL is MPC* The key points are:

A. Reinforcement learning (RL) and (approximate) dynamic programming (DP) address the same mathematical problem - optimal and suboptimal sequential decision making - and revolve around two central objects: value functions and policies.

Learning Objectives – Lecture 1, Part 1

- Understand MPC idea and basic problem formulation
- Collect impressions of what you can achieve in MPC applications

Outline

1. Optimization based Control

2. Concept of MPC

3. Applications

4. History of MPC

5. Summary

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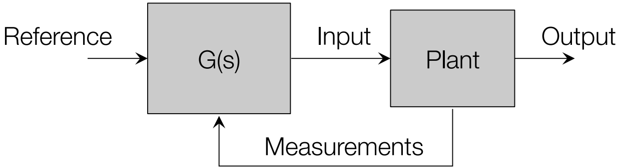
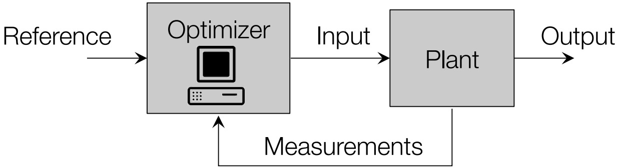
Optimization in the loop

Classical control loop:



The classical controller is replaced by an optimization algorithm:



The optimization uses predictions based on a model of the plant.

Optimization-based control: Motivation

Objective:

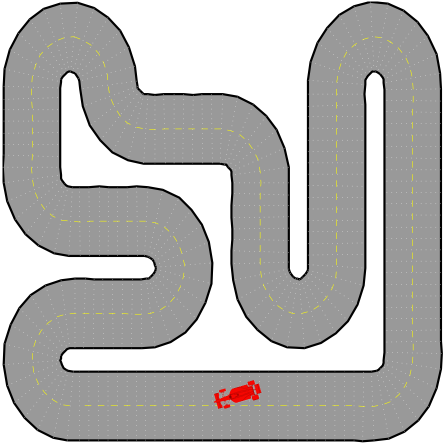
- Minimize lap time

Constraints:

- Avoid other cars
- Stay on road
- Don't skid
- Limited acceleration

Intuitive approach:

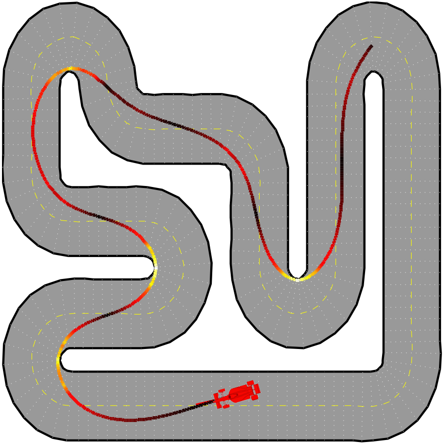
- Look forward and plan path based on
  - Road conditions
  - Upcoming corners
  - Abilities of car
  - etc...



Optimization-Based Control: Motivation

Minimize	(lap time)
while	avoid other cars
	stay on road
	...

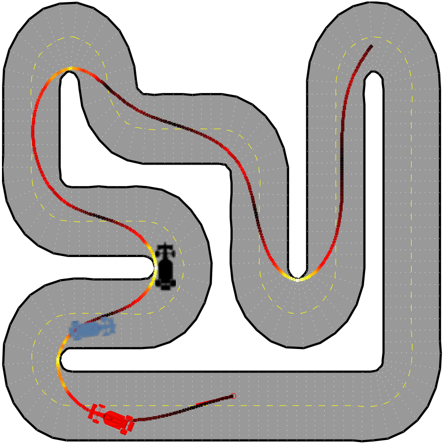
- Solve **optimization problem** to compute minimum-time path



Optimization-Based Control: Motivation

Minimize	(lap time)
while	avoid other cars
	stay on road
	...

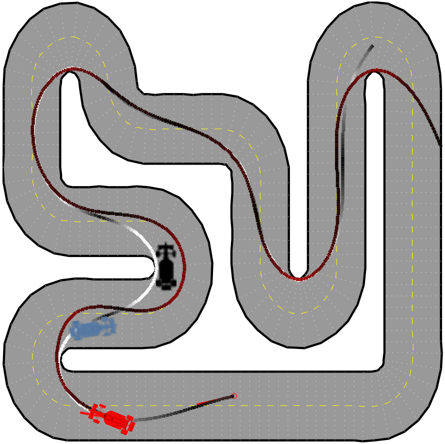
- Solve **optimization problem** to compute minimum-time path
- What to do if something unexpected happens?
  - We didn't see a car around the corner!
  - Must introduce **feedback**



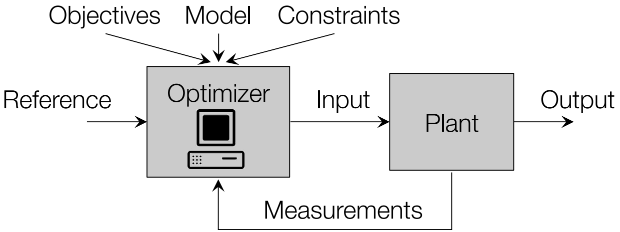
Optimization-Based Control: Motivation

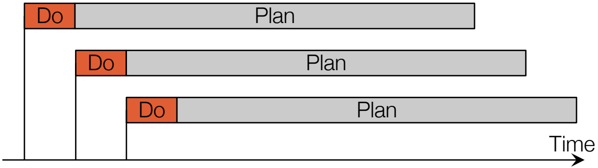
Minimize	(lap time)
while	avoid other cars
	stay on road
	...

- Solve **optimization problem** to compute minimum-time path
- Obtain series of planned control actions
- Apply **first** control action
- Repeat the planning procedure



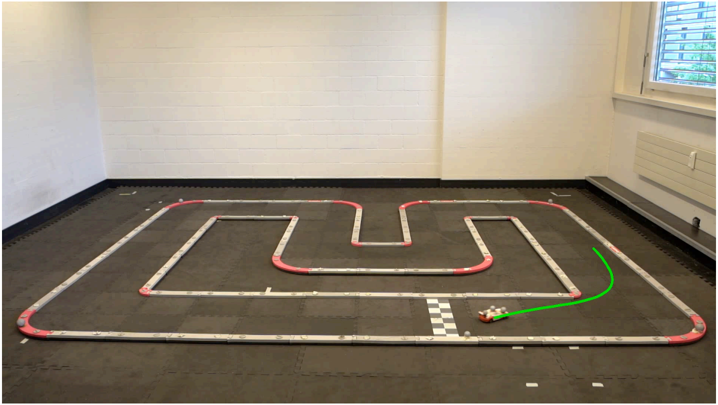
Model Predictive Control





Receding horizon strategy introduces **feedback**.

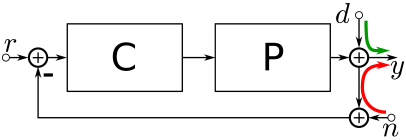
Illustration: MPC for Autonomous Racing



Outline

Two Different Perspectives

Classical design: design C

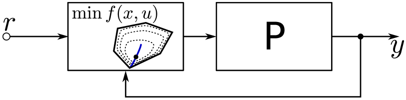


Dominant issues addressed

- Disturbance rejection ($d \to y$)
- Noise insensitivity ($n \to y$)
- Model uncertainty

(usually in **frequency domain**)

MPC: real-time, repeated optimization to choose $u(t)$ – often in supervisory mode



Dominant issues addressed

- Control constraints (limits)
- Process constraints (safety)

(usually in **time domain**)

Constraints in Control

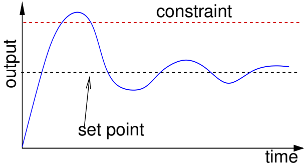
All physical systems have **constraints**:

- Physical constraints, e.g. actuator limits
- Performance constraints, e.g. overshoot
- Safety constraints, e.g. temperature/pressure limits

Optimal operating points are often near constraints.

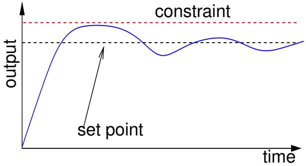
Classical control methods:

- Ad hoc constraint management
- Set point sufficiently far from constraints
- Suboptimal plant operation



Predictive control:

- Constraints included in the design
- Set point optimal
- Optimal plant operation



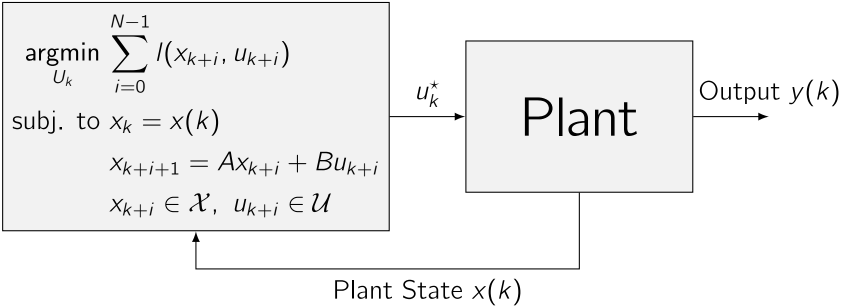
MPC: Mathematical Formulation

$$
\begin{aligned}
U_k^\star(x(k)) := \underset{U_k}{\operatorname{argmin}} \; & \sum_{i=0}^{N-1} l(x_{k+i}, u_{k+i}) & \\
\text{subj. to } \; & x_k = x(k) & \text{measurement} \\
& x_{k+i+1} = Ax_{k+i} + Bu_{k+i} & \text{system model} \\
& x_{k+i} \in \mathcal{X} & \text{state constraints} \\
& u_{k+i} \in \mathcal{U} & \text{input constraints} \\
& U_k = \{u_k, u_{k+1}, \ldots, u_{k+N-1}\} & \text{optimization variables}
\end{aligned}
$$

Problem is defined by

- **Objective** that is minimized
- Internal **system model** to predict system behavior
- **Constraints** that have to be satisfied

MPC: Mathematical Formulation



At each sample time:

- Measure / estimate current state $x(k)$
- Find the optimal input sequence for the entire planning window N: $U_k^\star = \{u_k^\star, u_{k+1}^\star, \dots, u_{k+N-1}^\star\}$
- Implement only the **first** control action $u_k^\star$

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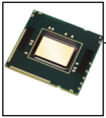
MPC: Applications



Computer control — ns

μs — Power systems

Traction control — ms

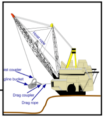
Seconds — Buildings

Refineries — Minutes

Hours — Nurse rostering

Train scheduling — Days

Weeks — Production planning



Outline

3. Applications

   Autonomous Race Cars

   Automotive Systems

   Path Following

   Autonomous Quadrocopter Flight

   Robotics

   Energy Efficient Building Control

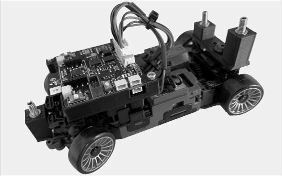
Autonomous Miniature Race Cars (Chronos)

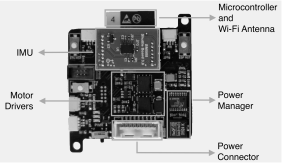
Race car:

- Based on widely available chassis
- 15cmx7cm, very light and fast (4m/s)
- Radio controlled

Control Problem:

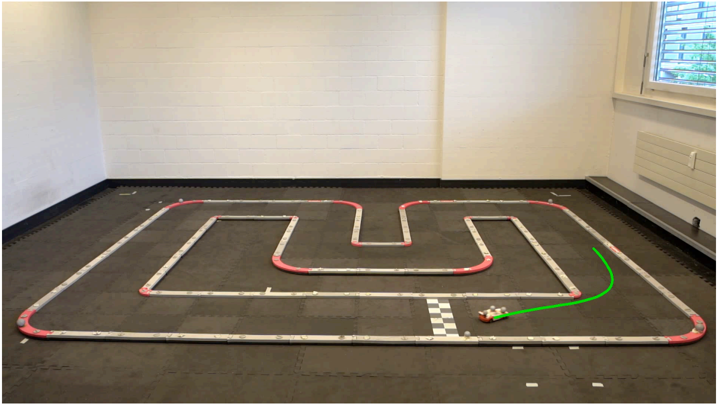
- **Nonlinear model** in 6D (position, orientation)
- **Constraints:** acceleration, steering angle, race track, other cars...
- **Task:** Optimal path planning and path following
- **Challenges:** State estimation, effects that are difficult to model/measure, e.g. slip, small sampling times





[Carron et al., ICRA 2023]

Autonomous Miniature Race Cars (Chronos)



Formula Student Driverless

[*Formula Student Driverless Competition 2018;* https://www.youtube.com/watch?v=2AAScaj_8t8]

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Ford Autonomous Driving on Ice

- Autonomous double-lane change.
- Road forecast and nonlinear vehicle model (driving on ice) used in MPC.
- MPC controls differential braking and steering.
- Experimental results @ 72 km/h on ice.

[*Falcone, Borrelli et al. International Journal Vehicle Autonomous Systems, 2009*]

Embotech Dynamic Motion Planning (w/ thyssenkrupp)

- Real-time (re-)computation of driving trajectory
- Avoidance of obstacles within physical limits of car/tires
- 70 kph, obstacle appears 17m in front of car. 60ms computation time.

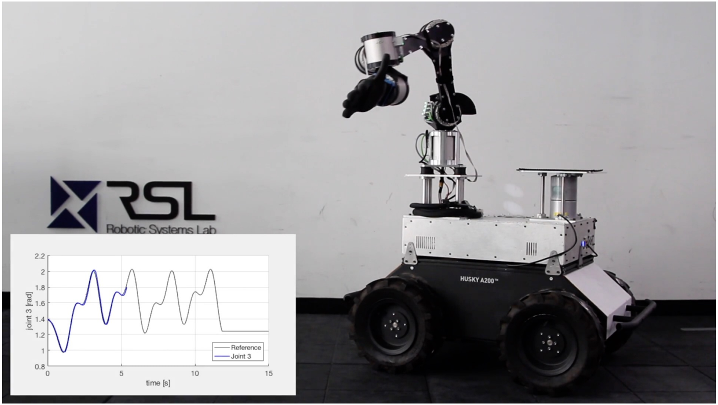
www.embotech.com/automotive

Outline

3. Applications

Path Following

MPC Control of a robot arm along a **known path**
Experiments on ANYpulator: 6-DoF robotic arm, series elastic actuators



[*Andrea Carron, IDSC/RSL, ETH Zurich*]

Outline

3. Applications

    Autonomous Race Cars

    Automotive Systems

    Path Following

    Autonomous Quadrocopter Flight

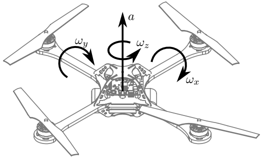
    Robotics

    Energy Efficient Building Control

Autonomous Quadrocopter Flight

Quadrocopters:

- Highly agile due to fast rotational dynamics
- High thrust-to-weight ratio allows for large translational accelerations
- Motion control by altering rotation rate and/or pitch of the rotors
- High thrust motors enable high performance control

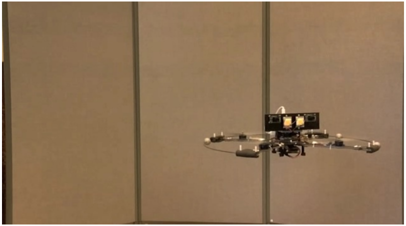


Control Problem:

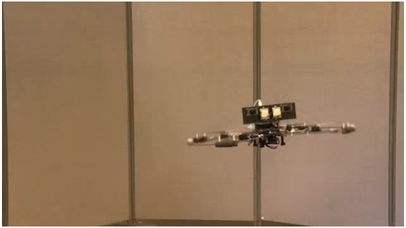
- **Nonlinear system** in 6D (position, attitude)
- **Constraints**: limited thrust, rates,...
- **Task**: Hovering, trajectory tracking
- **Challenges**: Fast unstable dynamics

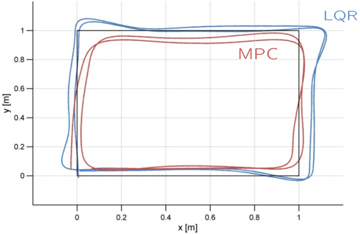
Autonomous Quadrocopter flight

LQR



MPC





[*M. Burri. Master Thesis ETH, 2011*]

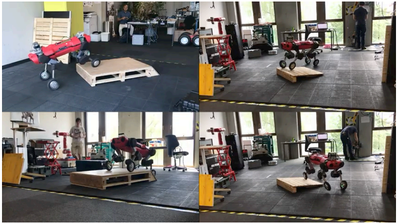
Model Predictive Contouring Control

[*Krinner et al., RSS 2024*; `https://www.youtube.com/watch?v=sbKe9emghtM`]

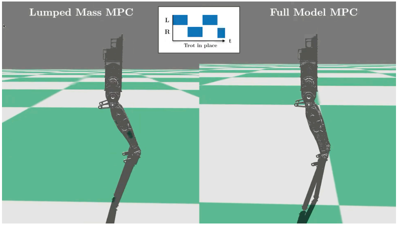
Outline

3. Applications

Examples: MPC in Robotics



[*Robotic Systems Lab. ETH Zurich;* https://www.youtube.com/watch?v=_rPvKlvyw2w]



[*AMBER Lab. Caltech;* https://www.youtube.com/watch?v=3g8ZNsCWdOA]

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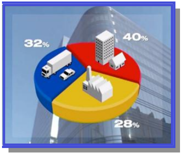
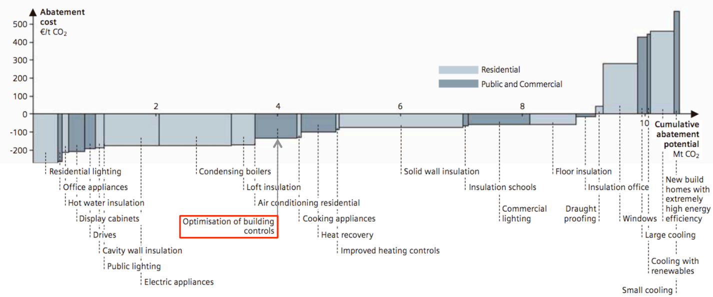
3. Applications
   - Autonomous Race Cars
   - Automotive Systems
   - Path Following
   - Autonomous Quadrocopter Flight
   - Robotics
   - Energy Efficient Building Control

Energy Efficient Building Control

- Buildings account for approx. **40% of global energy use**
- Most energy is consumed during use of the buildings
- Building sector has large potential for cost-effective reduction of CO_2 emissions
- Most investments in buildings are expected to pay back through **reduced energy bills**





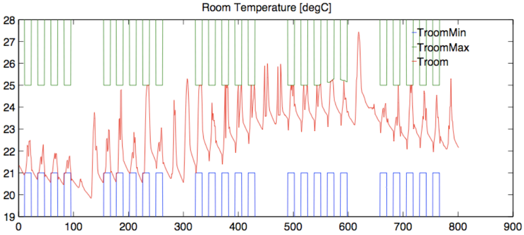
Greenhouse gas abatement cost curve for London buildings (2025, decision maker perspective)

Source: Watson, J. (ed.) (2008): Sustainable Urban Infrastructure, London Edition – a view to 2025. Siemens AG, Corporate Communications (CC) Munich, 71pp.

Energy Efficient Building Control

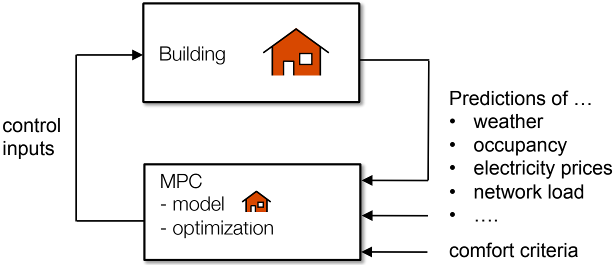
Integrated Room Automation:
Integrated control of heating, cooling, ventilation, electrical lighting, blinds,... of a single room/zone

Control Task: Use minimum amount of energy (or money) to keep room temperature, illuminance level and CO_2 concentration in **prescribed comfort ranges**



[*OptiControl Project, ETH. 2010;* `http://www.opticontrol.ethz.ch/`]

Energy Efficient Building Control



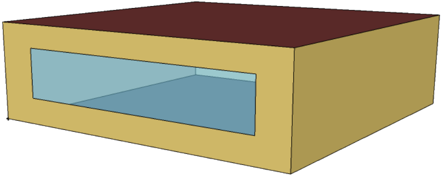
MPC opens the possibility to

- exploit building's **thermal storage capacity**
- use **predictions** of future disturbances, e.g. weather, for better planning
- use forecasts of electricity prices to shift electricity demand for grid-friendly behavior
- offer grid-balancing services to the power network
- ...

while respecting requirements for building usage (temperature, light, ...)

Example: One-Zone Building

Model of a one-zone building for climate control



- EnergyPlus model of a single zone
- Automatic extraction and linearization: openBuild tool
- Electric heating
- Weather: Jan, 2007 in San Francisco

$$x(k+1) = Ax(k) + Bu(k) + E_{rad}v_{rad}(k) + E_{amb}T_{amb}(k)$$

States		**Input**		**Disturbance**	
x_1	Zone temp.	u	Heat flux	v_{rad}	Solar radiation
$x_2..x_4$	Wall temps.			T_{amb}	Ambient temp.

Example: One-Zone Building

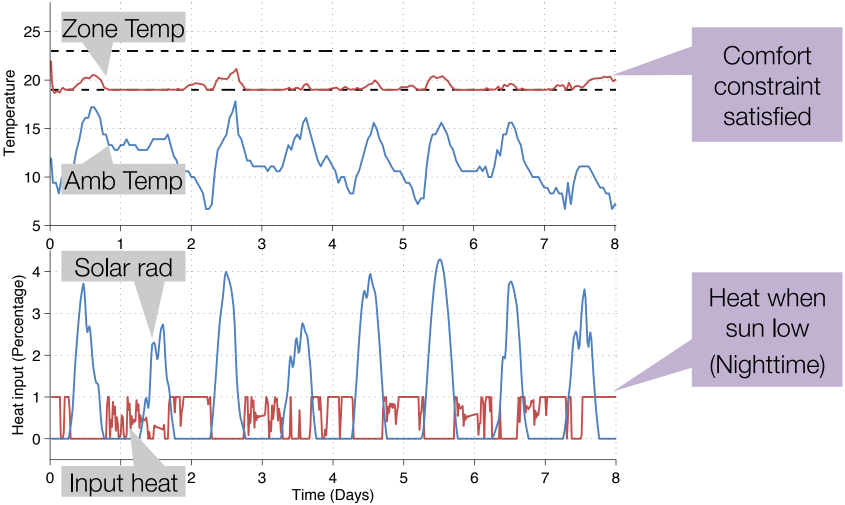
Problem formulation: Simplest Configuration

$$
\begin{aligned}
\min_{U_k} \; & \sum_{i=0}^{N-1} |u_{k+i}| \\
\text{subj. to } \; & x_k = x(k) \\
& x_{k+i+1} = Ax_{k+i} + Bu_{k+i} + E_{rad} v_{k+i,rad} + E_{amb} T_{k+i,amb} \\
& y_{k+i} = c^T x_{k+i} \\
& |y_{k+i} - 21| \leq 2 \\
& 0 \leq u_{k+i} \leq 1
\end{aligned}
$$

- Objective: Minimize total thermal energy use
- Forecast temperature / solar radiation N steps into the future
- Comfort constraints on room temperature
- Input: 0-100% heating

Example: One-Zone Building

Simple MPC



Annualized energy used: 81.6 kWh / m^2

Example: One-Zone Building

Problem formulation: Nighttime setbacks & pre-cooling

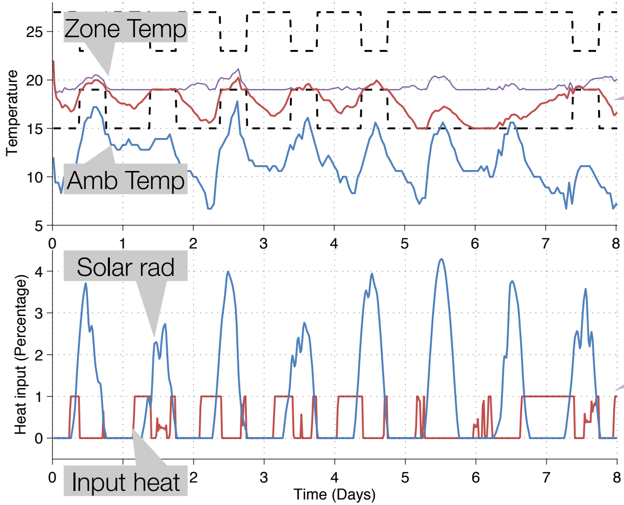
$$
\begin{aligned}
\min_{U_k} \ & \sum_{i=0}^{N-1} |u_{k+i}| \\
\text{subj. to } & x_k = x(k) \\
& x_{k+i+1} = Ax_{k+i} + Bu_{k+i} + E_{rad} v_{k+i,rad} + E_{amb} T_{k+i,amb} \\
& y_{k+i} = c^T x_{k+i} \\
& |y_{k+i} - 21| \leq 2 + \sigma_{k+i} \\
& 0 \leq u_{k+i} \leq 1
\end{aligned}
$$

Night Setbacks

$$
\sigma_{k+i} = \begin{cases} 0, & k+i \in \text{'daytime'} \\ 6, & k+i \in \text{'nighttime'} \end{cases}
$$

Example: One-Zone Building

Nighttime setback



Timing of setback is automatic

Early-morning pre-heating

Annualized energy used: 55.7 kWh / m^2

Example: One-Zone Building

Problem formulation: time-of-use pricing

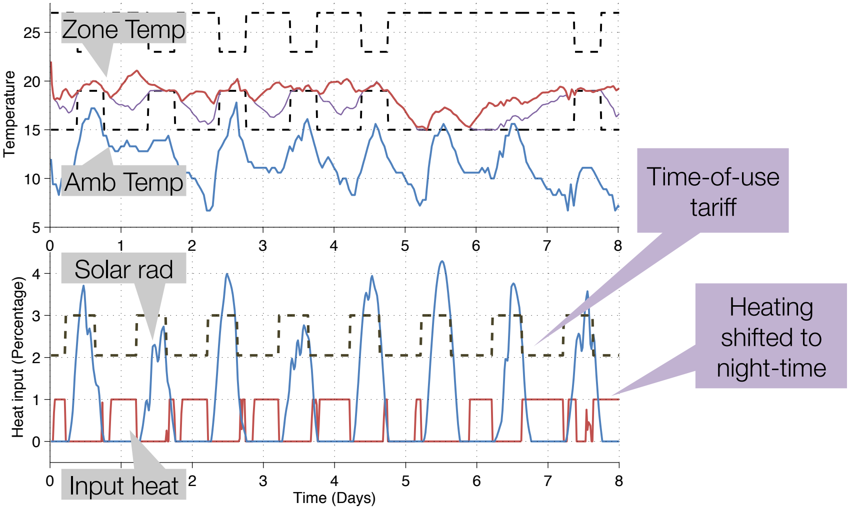
$$
\begin{aligned}
\min_{U_k} \; & \sum_{i=0}^{N-1} c_{k+i} \cdot |u_{k+i}| \\
\text{subj. to } & x_k = x(k) \\
& x_{k+i+1} = Ax_{k+i} + Bu_{k+i} + E_{rad} v_{k+i,rad} + E_{amb} T_{k+i,amb} \\
& y_{k+i} = c^T x_{k+i} \\
& |y_{k+i} - 21| \leq 2 + \sigma_{k+i} \\
& 0 \leq u_{k+i} \leq 1
\end{aligned}
$$

Time-of-Use Tariff

$$
c_{k+i} = \begin{cases} c_{day} & \text{at daytime 9h – 18h} \\ c_{night} & \text{at nightime 18h – 9h} \end{cases}
$$

Example: One-Zone Building

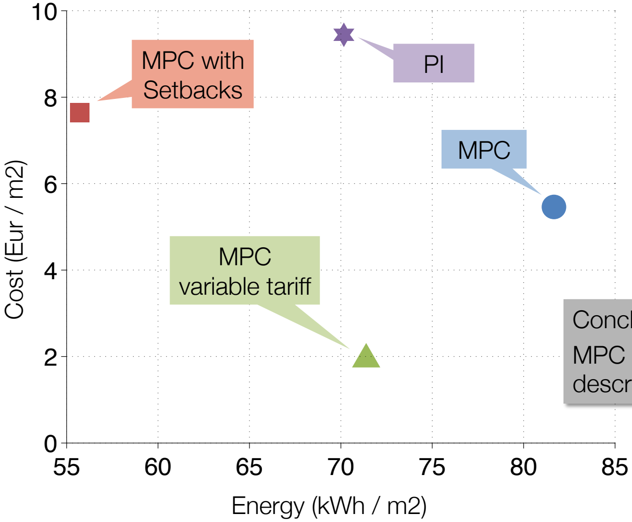
Time-of-use tariff



Annualized energy used: 71.3 kWh / m^2

Example: One-Zone Building

Annualized Comparison



Conclusion:

MPC is easy to tune / describe complex goals

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1. Optimization based Control

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3. Applications

4. History of MPC

5. Summary

History of MPC

- **A. I. Propoi, 1963**, "Use of linear programming methods for synthesizing sampled-data automatic systems", **Automation and Remote Control**.

- **J. Richalet et al., 1978** "Model predictive heuristic control- application to industrial processes". **Automatica**, 14:413-428.
  - known as **IDCOM (Identification and Command)**
  - impulse response model for the plant, linear in inputs or internal variables (**only stable plants**)
  - quadratic performance objective over a finite prediction horizon
  - future plant output behavior specified by a reference trajectory
  - **ad hoc** input and output constraints
  - optimal inputs computed using a heuristic iterative algorithm, interpreted as the dual of identification
  - controller was not a transfer function, hence called **heuristic**

History of MPC

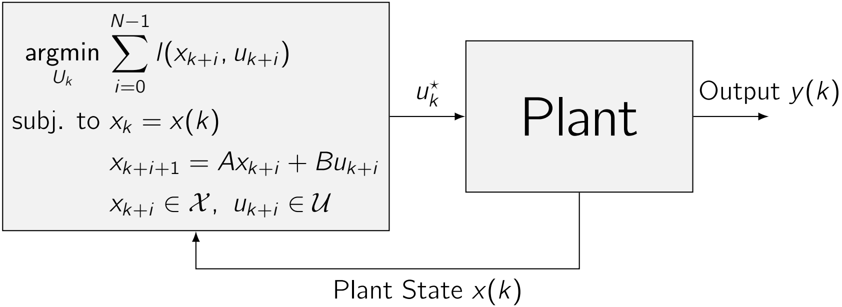
- 1970s: Cutler suggested MPC in his PhD proposal at the University of Houston in 1969 and introduced it later at Shell under the name Dynamic Matrix Control. **C. R. Cutler, B. L. Ramaker, 1979** "Dynamic matrix control – a computer control algorithm". **AICHE National Meeting**, Houston, TX.
  - successful in the petro-chemical industry
  - linear step response model for the plant
  - quadratic performance objective over a finite prediction horizon
  - future plant output behavior specified by trying to follow the set-point as closely as possible
  - input and output constraints included in the formulation
  - optimal inputs computed as the solution to a least–squares problem
  - **ad hoc** input and output constraints. Additional equation added online to account for constraints. Hence a **dynamic matrix** in the least squares problem.
- **C. Cutler, A. Morshedi, J. Haydel, 1983**. "An industrial perspective on advanced control". **AICHE Annual Meeting**, Washington, DC.
  - Standard QP problem formulated in order to systematically account for constraints.

History of MPC

- Mid 1990s: extensive theoretical effort devoted to provide conditions for guaranteeing feasibility and closed-loop stability
- 2000s: development of tractable robust MPC approaches; nonlinear and hybrid MPC; MPC for very fast systems
- 2010s: stochastic MPC; distributed large-scale MPC; economic MPC
- 2020s: learning-based MPC

Outline

Summary: MPC



At each sample time:

- Measure /estimate current state $x(k)$
- Find the **optimal input sequence** for the entire planning window N:
  $U_k^\star = \{u_k^\star, u_{k+1}^\star, \ldots, u_{k+N-1}^\star\}$
- Implement only the **first** control action $u_k^\star$

Important Aspects of Model Predictive Control

Main advantages:

- Systematic approach for handling **constraints**
- High **performance** controller

Main challenges:

- **Implementation**
  MPC problem has to be solved in real-time, i.e. within the sampling interval of the system, and with available hardware (storage, processor,...).
- **Feasibility**
  Optimization problem may become infeasible at some future time step, i.e. there may not exist a plan satisfying all constraints
- **Stability**
  Closed-loop stability, i.e. convergence, is not automatically guaranteed
- **Robustness**
  The closed-loop system is not necessarily robust against uncertainties or disturbances